

SESSION 7

Task 1: Process Synchronization and Child Process Monitoring

Task:

Implement process synchronization and child-process monitoring using `waitpid()`

```
root@HP:~# nano session7_waitpid.c
root@HP:~# gcc session7_waitpid.c -o session7_waitpid
root@HP:~# gcc session7_waitpid.c -o session7_waitpid
root@HP:~# ./session7_waitpid
Parent: creating child process...
Parent: waiting for child using waitpid()...
Child: running for 3 seconds...
Child: finished.
Parent: child exited normally.
Parent: child exit status = 42
Parent: done.
root@HP:~# |
```

```
root@HP:~# nano session7_path.c
root@HP:~# gcc session7_path.c -o session7_path
root@HP:~# ./session7_path
Searching PATH for 'ls'...

Checking: /usr/local/sbin/ls
Checking: /usr/local/bin/ls
Checking: /usr/sbin/ls
Checking: /usr/bin/ls

Command found!
Executable: /usr/bin/ls
root@HP:~# |
```

Task 2: PATH Variable and Command Resolution

Task:

Retrieve the `PATH` variable, parse search directories, locate executables, verify permissions, handle missing commands, and test command resolution.

```

root@HP:~# ./session7_path_exec
Found command: /usr/bin/ls
Starting child process...
total 424
-rw-r--r-- 1 root root      0 Aug  2 05:29 demo1.c
-rwxr-xr-x 1 root root 16088 Aug  2 05:44 demo2
-rw-r--r-- 1 root root   516 Aug  2 05:44 demo2.c
-rwxr-xr-x 1 root root 16280 Aug  1 04:16 fork_example
-rw-r--r-- 1 root root   591 Aug  1 04:15 fork_example.c
-rw----- 1 root root    27 Aug  1 03:52 nano.867.save
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neeha
drwxr-xr-x 2 root root  4096 Jul 31 18:23 neetu
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 process
-rw-r--r-- 1 root root   706 Aug  8 03:16 process.c
-rwxr-xr-x 1 root root 16128 Aug 20 09:59 session1
-rw-r--r-- 1 root root   441 Aug 20 09:58 session1.c
-rwxr-xr-x 1 root root 16016 Aug 20 10:13 session1_exec
-rw-r--r-- 1 root root   215 Aug 20 10:13 session1_exec.c
-rwxr-xr-x 1 root root 16184 Aug 20 10:16 session2
-rw-r--r-- 1 root root   349 Aug 20 10:10 session2.c
-rwxr-xr-x 1 root root 16200 Aug 20 10:19 session2_keyboard
-rw-r--r-- 1 root root   627 Aug 20 10:19 session2_keyboard.c
-rwxr-xr-x 1 root root 16320 Aug 20 14:13 session3_history
-rw-r--r-- 1 root root  1227 Aug 20 14:13 session3_history.c
-rwxr-xr-x 1 root root 16184 Aug 20 14:15 session3_memory
-rw-r--r-- 1 root root  1030 Aug 20 14:15 session3_memory.c
-rwxr-xr-x 1 root root 16232 Aug 20 14:25 session4_parser
-rw-r--r-- 1 root root   1193 Aug 20 14:25 session4_parser.c
-rwxr-xr-x 1 root root 16200 Aug 20 14:23 session4_tokenizer
-rw-r--r-- 1 root root   611 Aug 20 14:23 session4_tokenizer.c
-rwxr-xr-x 1 root root 16280 Aug 20 14:27 session5_quotes
-rw-r--r-- 1 root root  1258 Aug 20 14:27 session5_quotes.c
-rwxr-xr-x 1 root root 16056 Aug 20 14:37 session6_error
-rw-r--r-- 1 root root   231 Aug 20 14:37 session6_error.c
-rwxr-xr-x 1 root root 15968 Aug 20 14:34 session6_escape
-rw-r--r-- 1 root root   291 Aug 20 14:34 session6_escape.c
-rwxr-xr-x 1 root root 16056 Aug 20 14:37 session6_exec_date
-rw-r--r-- 1 root root   164 Aug 20 14:36 session6_exec_date.c
-rwxr-xr-x 1 root root 16144 Aug 20 14:35 session6_execute
-rw-r--r-- 1 root root   545 Aug 20 14:35 session6_execute.c
-rwxr-xr-x 1 root root 16264 Aug 20 14:41 session7_path
-rw-r--r-- 1 root root   931 Aug 20 14:41 session7_path.c
-rwxr-xr-x 1 root root 16440 Aug 20 14:43 session7_path_exec
-rw-r--r-- 1 root root  1358 Aug 20 14:42 session7_path_exec.c
-rwxr-xr-x 1 root root 16232 Aug 20 14:40 session7_waitpid
-rw-r--r-- 1 root root   719 Aug 20 14:39 session7_waitpid.c
drwxr-xr-x 3 root root  4096 Aug 20 09:57 shellforge
drwxr-xr-x 2 root root  4096 Jul 31 18:16 sri
-rwxr-xr-x 1 root root 16136 Aug  8 03:18 states
-rw-r--r-- 1 root root   576 Aug  8 03:17 states.c
-rw-r--r-- 1 root root      0 Jul 31 18:28 xls.txt

Child finished with exit status: 0
root@HP:~# |

```

```

root@HP:~# ./session7_path
Searching PATH for 'abcdef123'...

Checking: /usr/local/sbin/abcdef123
Checking: /usr/local/bin/abcdef123
Checking: /usr/sbin/abcdef123
Checking: /usr/bin/abcdef123
Checking: /sbin/abcdef123
Checking: /bin/abcdef123
Checking: /usr/games/abcdef123
Checking: /usr/local/games/abcdef123
Checking: /usr/lib/wsl/lib/abcdef123
Checking: /mnt/c/Program Files/Common Files/Oracle/Java/javapath/abcdef123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8path/abcdef123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath/abcdef123
Checking: /mnt/c/WINDOWS/system32/abcdef123
Checking: /mnt/c/WINDOWS/abcdef123
Checking: /mnt/c/WINDOWS/System32/Wbem/abcdef123
Checking: /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0//abcdef123
Checking: /mnt/c/WINDOWS/System32/OpenSSH//abcdef123
Checking: /mnt/c/Program Files/dotnet//abcdef123
Checking: /mnt/c/Program Files/nodejs//abcdef123
Checking: /mnt/c/ProgramData/chocolatey/bin/abcdef123
Checking: /mnt/c/Program Files/Microsoft SQL Server/170/Tools/Binn//abcdef123
Checking: /mnt/c/Program Files/Microsoft SQL Server/Client SDK/ODBC/170/Tools/Binn/
/abcdef123
Checking: /mnt/c/Users/shahb/AppData/Local/Programs/Python/Python313/Scripts//abcde
f123
Checking: /mnt/c/Users/shahb/AppData/Local/Programs/Python/Python313//abcdef123
Checking: /mnt/c/Users/shahb/AppData/Local/Programs/Python/Launcher//abcdef123
Checking: /mnt/c/Users/shahb/AppData/Local/Microsoft/WindowsApps/abcdef123
Checking: /mnt/d/ffmpeg-master-latest-win64-gpl/bin/abcdef123
Checking: /mnt/c/Users/shahb/AppData/Roaming/npm/abcdef123
Checking: /mnt/c/Users/shahb/AppData/Local/Python/bin/abcdef123
Checking: /mnt/c/Users/shahb/AppData/Local/Kingsoft/WPS Office/12.1.0.27458/clitool
/abcdef123
Checking: /mnt/d/vs vsvsvsvs/bin/abcdef123
Checking: /snap/bin/abcdef123

Command not found in PATH.

```

Overall Observation – Session 7

Session 7 focused on process synchronization and command resolution. The first task provided practical experience with monitoring child processes and synchronizing their execution using `waitpid()`. The second task demonstrated how the shell retrieves and processes the `PATH` environment variable to locate executable commands. Together, the activities provided a practical understanding of both process coordination and how a shell resolves commands before execution.

SESSION 8

Task 1: Variable Detection and Expansion

Task:

Detect variable references, expand values, handle undefined variables, update tokens, support nested variables, and test expansion logic.

```
root@HP:~# nano session8.c
root@HP:~# gcc session8.c -o session8
root@HP:~# ./session8
ShellForge - Session 8
=====
myshell>
root@HP:~# nano session8.c
root@HP:~# gcc session8.c -o session8
root@HP:~# ./session8
ShellForge - Session 8
=====
myshell> |
=====
myshell> set NAME Neeharika
Input: set NAME Neeharika
Expanded: set NAME Neeharika
Variable NAME set.
myshell> echo $NAME
Input: echo $NAME
Expanded: echo Neeharika
Neeharika
myshell> echo ${NAME}
Input: echo ${NAME}
Expanded: echo Neeharika
Neeharika
myshell> echo $DOES_NOT_EXIST
Input: echo $DOES_NOT_EXIST
Expanded: echo
myshell> |
myshell> echo Hello ShellForge
Input: echo Hello ShellForge
Expanded: echo Hello ShellForge
Hello ShellForge
myshell> echo My name is $NAME
Input: echo My name is $NAME
Expanded: echo My name is Neeharika
My name is Neeharika
myshell> |
```

Task 2: Built-in Commands and Dispatch Table

Task:

Identify built-ins, create a dispatch table, execute in-process commands, handle invalid commands, maintain state, and test dispatch logic.

```
myshell> pwd
Input: pwd
Expanded: pwd
/root
myshell> cd ..
Input: cd ..
Expanded: cd ..
myshell> pwd
Input: pwd
Expanded: pwd
/
myshell>
myshell> history
Input: history
Expanded: history
1  set NAME Neeharika
2  echo $NAME
3  echo ${NAME}
4  echo $DOES_NOT_EXIST
5  echo Hello ShellForge
6  echo My name is $NAME
7  pwd
8  cd ..
9  pwd
10 history
myshell>
myshell> ls
Input: ls
Expanded: ls
External command found: /usr/bin/ls
bin      etc      lib.usr-is-merged  mnt      run      srv      var
bin.usr-is-merged  home    lib64             opt      sbin     sys
boot      init    lost+found        proc     sbin.usr-is-merged  tmp
dev        lib     media             root     snap     usr
myshell> |
```

```
myshell> abcdef123
Input: abcdef123
Expanded: abcdef123
abcdef123: command not found
myshell> |

myshell> exit
Input: exit
Expanded: exit
root@HP:~# |
```

Overall Observation – Session 8

Session 8 focused on variable expansion and the management of built-in shell commands. The first task provided practical experience in detecting variable references, expanding values, and handling undefined and nested variables. The second task focused on identifying built-in commands and using a dispatch table to execute them while maintaining shell state. Overall, the session improved the understanding of how a shell interprets variables and manages commands internally.